

08 - Guideline Logic and Practice Tension

Companion reading handout for simultaneous listening.

Course: Medical Prevention of Spontaneous Early Delivery

Guidelines are not commandments floating above evidence.

They are structured interpretations of evidence inside a clinical system. A good resident should know the recommendation, but also the evidence boundary that gives the recommendation its shape.

SMFM

Start with Society for Maternal-Fetal Medicine Consult Series number seventy.

For an asymptomatic singleton pregnancy without a prior spontaneous preterm birth, if the transvaginal cervical length is twenty millimeters or less before twenty-four weeks, SMFM recommends vaginal progesterone. If the cervical length is twenty-one to twenty-five millimeters, SMFM recommends considering vaginal progesterone through shared decision-making.

That is not two ways of saying the same thing.

Twenty millimeters or less is the stronger recommendation zone. Twenty-one to twenty-five millimeters is a zone of lower certainty, lower absolute risk on average, and more preference-sensitive decision-making. Twenty-five millimeters diagnoses the short cervix. Twenty millimeters or less is where the recommendation becomes strongest. The memory is a gradient of certainty, not simply a short-cervix rule.

Now compare ACOG's updated progestogen guidance.

ACOG addressed the withdrawal of seventeen alpha hydroxyprogesterone caproate. It states that intramuscular 17-OHPC is not recommended for primary prevention of preterm birth in patients with a history of spontaneous preterm birth. It also states that vaginal progesterone may be considered in patients with a history of preterm birth, singleton gestation, and shortened cervix, but it has not been proven effective in the absence of a shortened cervix and should not be considered an alternative to 17-OHPC.

Again, the population boundary is the teaching point.

Prior spontaneous preterm birth alone is not the same as current short cervix. Current short cervix alone is not the same as prior spontaneous preterm birth. The

overlap may create a higher-risk phenotype, but it does not erase the need to identify which evidence applies.

The evidence behind that guidance is not merely regulatory language. PROLONG failed to confirm the 17-OHPC benefit in prior spontaneous preterm birth. Conde-Agudelo and Romero then showed why vaginal progesterone should not be casually substituted in prior preterm birth without a short cervix: the apparent benefit in the whole meta-analysis was carried by small studies, while the large studies were essentially null, with relative risks near one for birth before thirty-seven and before thirty-four weeks.

Late Short Cervix

Now think about late short cervix.

Some clinical environments encounter patients whose cervix is first found short after twenty-four weeks. The temptation is understandable. If vaginal progesterone helps when the cervix is short before twenty-four weeks, perhaps it helps whenever the cervix is short.

But guidelines are cautious because the evidence is not the same.

The strongest randomized short-cervix evidence lives in the mid-trimester, before or around twenty-four weeks. The late-initiation evidence is mostly observational. In low-risk or no-prior-preterm-birth patients, several late-initiation studies do not show benefit. Dominsky found no reduction in spontaneous preterm birth thresholds and no prolongation of diagnosis-to-delivery interval. The two thousand twenty-six AJOG MFM abstract likewise found no time-to-delivery benefit in asymptomatic women first diagnosed after twenty-four weeks. Einum showed that the apparent benefit diminished with later diagnosis, with no apparent benefit after twenty-eight weeks.

The high-risk prior-preterm-birth late-short-cervix data are more suggestive but not definitive. Luxembourg reported lower crude rates in a high-risk population, but the adjusted hazard ratio for delivery before thirty-seven weeks was not statistically significant and the design was retrospective.

This is how a guideline-minded physician should speak:

"The evidence is strongest for starting vaginal progesterone before twenty-four weeks in an asymptomatic singleton short-cervix phenotype. After twenty-four weeks, especially after twenty-eight weeks, we are no longer in the same evidence zone. The decision may be individualized, but we should not pretend that the randomized mid-trimester data simply extend unchanged."

Notice what that statement does. It does not forbid thought. It prevents false certainty.

Shared Decision-Making

Shared decision-making deserves the same discipline.

Shared decision-making does not mean, "Evidence is vague, so anything is fine." It means the clinician explicitly shares the population match, likely benefit, uncertainty, burden, safety concerns, and alternatives. In this course, a high-quality shared decision-making conversation might say:

"Your cervix is in a range where risk is higher, but the strongest recommendation is at twenty millimeters or less before twenty-four weeks. At twenty-two millimeters, the evidence is less absolute. Vaginal progesterone is low burden for many patients, but the expected benefit is less certain than in the very short-cervix trials."

For late initiation, it might say:

"Because the cervix was first found short after twenty-four weeks, the evidence for starting progesterone is weaker. Some observational studies do not show longer pregnancy duration. If we consider treatment, we should call it extrapolated or uncertain, not proven."

That is what adult medical counseling sounds like. It is precise, not evasive.

Local Practice Versus Evidence

The resident should also understand why local practice may differ from a strict guideline reading. Local practice may be influenced by medication availability, patient anxiety, clinician experience, low perceived harm, national screening policy, or institutional culture. Those are real forces. But none of them should be confused with strong efficacy evidence.

Safety is not efficacy. A low-risk treatment can still be ineffective. And an ineffective treatment can still create cost, false reassurance, medicalization, and distraction from other management categories.

So the guideline memory is this:

SMFM: asymptomatic singleton, no prior spontaneous preterm birth, twenty millimeters or less before twenty-four weeks, recommend vaginal progesterone. Twenty-one to twenty-five millimeters, discuss.

ACOG: 17-OHPC is not recommended after the regulatory evidence review; vaginal progesterone is not a substitute for prior preterm birth without shortened cervix.

Late initiation: do not borrow the confidence of the pre-twenty-four-week trials.

Guidelines should not make the physician passive. They should make the physician honest about where the evidence is strong, where it is conditional, and where practice becomes extrapolation.